

Fig. 2.9 Fluorescent tube

The induced EMF is nearly about 1200V. This voltage is applied between the filaments of the tube light, causing electron flow between the filament and the inert gas of the tube light. The electron collision in to the inert gas produces the ultra violet rays. These rays impinge on the phosphorous coating. Light is emitted by the coating. After light started at 110V is enough for the light to retain the supply voltage 120 volt, which is dropped across the choke. The luminous efficiency of a fluorescent lamp is 60 lumens / watt.

b. Advantages

1. Voltage fluctuation has very small effect on light output.
2. The luminous efficiency is more as length of rod is more.
3. It gives light close to natural light.
4. Heat radiations are negligible.

c. Disadvantages

1. Its brightness is less.
2. Initial cost is more
3. Overall maintenance cost is high.

d. Applications

1. Fluorescent lamp is available in required designs and sizes. Hence it is used largely in residential areas.
2. It is used for good lighting.
3. Special fluorescent lights are used in stage lighting for films and in video camera lighting.

2.7.2 Compact Fluorescent Lamp (CFL)

The compact fluorescent lamps are becoming very useful nowadays, because of consumption of power, cost, longer life, attractive look, smooth light and low maintenance. These lamps are available in different sizes and designs. They have single rod, double rod, triple rod or spiral rod. These lamps are available in different power ratings like 5, 7, 9, 11, 18 and 24 watts in 220 V. It is shown in figure 2.10.



Fig. 2.10 Compact Fluorescent Lamp (CFL)

It is basically a low pressure mercury vapour lamp having two electrodes coated with an electron-emitting material placed in a glass tube. The tube is coated internally with some fluorescent material in the form of powder. In the tube one drop of mercury and argon gas is filled at low pressure. Compact fluorescent lamps are now available in the same popular sizes as incandescent and are used as an energy-saving alternative in homes

a. Advantages

1. Low energy consumption.
2. Low maintenance cost
3. It starts instantly
4. It does not heat the surroundings
5. Excellent colour properties

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